

Research Area

1. Basic Research

- **Focus:** Fundamental understanding of diseases, immune responses, or pathogens not directed toward a specific product.
 - **Stages:** Technology Readiness Levels (TRL) 1-3.
 - **Includes:**
 - Studies on pathogen mechanisms, transmission, virulence, immunology, genetics, microbiome roles, and One Health interactions.
 - Development of models and technologies assisting in drug and vaccine design and testing.
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2. Therapeutics

- **Focus:** Treatment of infections using antimicrobials.
 - **Includes:**
 - Improvement of existing antimicrobials and therapies.
 - Combination therapies and dose optimization.
 - Development of new antimicrobials and alternatives (e.g., antibodies, probiotics, bacteriophages).
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3. Vaccines

- **Focus:** Products stimulating the adaptive immune system to prevent disease.
 - **Includes:**
 - Research on vaccine development challenges (e.g., antigen identification, delivery methods).
 - Identification and assessment of vaccine candidates for safety and efficacy.
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4. Preventives - Other

- **Focus:** Non-vaccine products preventing disease without inducing specific immune memory.
 - **Includes:**
 - Prophylactic treatments.
 - Immune modulators enhancing general immune function.
 - Breeding for disease resistance (e.g., genome editing).
 - Anti-biofilm agents.
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5. Diagnostics

- **Focus:** Development or improvement of detection, screening, or diagnostic tools.
- **Includes:**
 - Identification of causative agents and resistance profiles.
 - Diagnostic tests for clinical and field use.
 - Tools for surveillance and epidemiological studies.
 - Companion diagnostics and machine learning detection tools.

6. Promotants

- **Focus:** Products improving health or productivity in the absence of disease.
- **Includes:**
 - Probiotics and prebiotics.
 - Antimicrobial peptides and phytochemicals.
 - Organic acids, enzymes, and other growth-promoting alternatives.

7. Other Products

- **Focus:** External or topical applications not classified elsewhere.
- **Includes:**
 - Biocides (antiseptics, disinfectants).
 - Biofilm prevention materials.
 - Medical devices and wound care products.
 - Health monitoring technologies (e.g., sensors in animals).

8. Operational Research

- **Focus:** Aiding decision-making and management strategies.
- **Includes:**
 - Infection prevention and control practices.
 - Antimicrobial stewardship and optimal use studies.
 - Access and availability improvements for AMR technologies.
 - Surveillance and epidemiological studies.
 - Social science research on behavior change and economic analyses.

9. Capacity Building

- **Focus:** Enhancing human and infrastructural resources to tackle AMR.
- **Includes:**
 - Laboratory capacity enhancement.
 - Training and staff development.
 - Network formation for knowledge sharing.
 - Process improvements beyond single products.

10. Policy

- **Focus:** Informing policy and regulation development or revision.
- **Includes:**
 - Research proposing policy changes or stakeholder action.
 - Studies on healthcare services impact, economic models, and health technology assessments.

Product Development Stages

1. Discovery

- **TRL:** 4-5.
 - **Includes:**
 - Target validation and hit discovery for therapeutics and preventives.
 - Vaccine component identification and improvement.
 - Diagnostic concept development and prototype creation.
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2. Development

- **TRL:** 6-8.
 - **Includes:**
 - Clinical trials (Phases 1-3) for human therapeutics and preventives.
 - Safety and effectiveness studies for animal health products.
 - Validation of manufacturing processes and product performance.
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3. Approval and Post-Approval

- **TRL:** 9.
- **Includes:**
 - Regulatory filings in various jurisdictions.
 - Post-approval obligations (e.g., pharmacovigilance).
 - Product optimization research.
 - Clinical studies informing product use and guidelines.

Technology Readiness Levels (TRL)

- TRL 1: Basic principles observed.
- TRL 2: Concept formulated.
- TRL 3: Experimental proof of concept.
- TRL 4: Technology validated in the lab.
- TRL 5: Technology validated in a relevant environment.
- TRL 6: Technology demonstrated in a relevant environment.
- TRL 7: System prototype in an operational environment.
- TRL 8: System complete and qualified.
- TRL 9: Actual system proven in operational environment.